OR-AS Operations Research Applications and Solutions	Case Name: Residential House Finishing Works (1)	Sector	**Construction (residential building)**
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	William & Jef		User defined distributions
Date	2016	Project Control	Automatic tracking
File Name	C2016-16 Residential House Finishing Works (1)		Tracking based on user input

1. Project description

Project authenticity

Residential House Finishing Works (1) is a project focused on completing the final stages of a residential building through a structured, multi-phase approach. It includes the installation and finishing of essential systems such as ventilation, sanitation, electrical works, insulation, flooring, plastering, and interior elements like kitchens, stairs, and carpentry, followed by final detailing to ensure the house is fully functional, comfortable, and ready for occupancy.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	24
Planned Duration (PD)	90d
Budget At Completion (BAC)	54,577.76 €
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	69%
Activity Distribution (AD)	68%
Length of Arcs (LA)	50%
Topological Float (TF)	28%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	100	0	N/A
CRI-tau	100	0	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	71	37	-0.84
SI	84	29	-2.12
SSI	17	10	0.15
CRI-r	19	13	0.41
CRI-rho	19	13	0.33
CRI-tau	11	8	0.57

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	2.32%	0.91%
PV - SPI	4.47%	1.12%
PV - SCI	4.47%	1.12%
ED - 1	2.73%	1.14%
ED - SPI	4.43%	1.07%
ED - SCI	4.43%	1.07%
ES - 1	2.10%	-0.94%
ES - SPI(t)	9.29%	-7.49%
ES - SCI(t)	9.29%	-7.49%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0	0.00%
CPI	0	0.00%
SPI	2.22%	0.17%
SPI(t)	8.07%	-7.09%
SCI	2.22%	0.17%
SCI(t)	8.07%	-7.09%
0.8 CPI + 0.2 SPI	0.46%	0.16%
0.8 CPI + 0.2 SPI(t)	1.22%	-1.01%

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-6581.75	0.00	0.00	0.81	1.00	1.00	1.00
std dev	3086.35	0.00	0.00	0.03	0.00	0.00	0.00
final	-9948.99	0.00	0.00	0.85	1.00	1.00	1.00

2.3.3.2. Time forecasting

PD	90d	Real Duration	90d	Late/early	0%

EAC(t)			**Real Accuracy**	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	90.00	0.00	0.00%	0.00%
PV - SPI	90.00	0.00	0.00%	0.00%
PV - SCI	112.25	4.57	24.72%	-24.72%
ED - 1	90.00	0.00	0.00%	0.00%
ED - SPI	90.00	0.00	0.00%	0.00%
ED - SCI	97.75	8.38	8.61%	-8.61%
ES - 1	90.00	0.00	0.00%	0.00%
ES - SPI(t)	90.00	0.00	0.00%	0.00%
ES - SCI(t)	97.75	8.38	8.61%	-8.61%

2.3.3.3. Cost forecasting

BAC	54,577.76 €	Real Cost	64,526.76 €	Over/under budget	18%

EAC			**Real Accuracy**	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	61159.51	3086.35	5.22%	5.22%
CPI	67818.62	2898.89	5.10%	-5.10%
SPI	61159.51	3086.35	5.22%	5.22%
SPI(t)	61159.51	3086.35	5.22%	5.22%
SCI	67818.62	2898.89	5.10%	-5.10%
SCI(t)	67818.62	2898.89	5.10%	-5.10%
0.8 CPI + 0.2 SPI	66206.07	1427.04	2.60%	-2.60%
0.8 CPI + 0.2 SPI(t)	66206.07	1427.04	2.60%	-2.60%